

Persona Vectors as a Psychometric Instrument for Moral Character in Language Models

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We apply the persona vector method of Chen et al. (2025) to four moral character traits — **honesty**, **harmlessness**, **fairness**, and **compassion** — in Gemma-3-12B-IT, and conduct the first systematic psychometric evaluation of the resulting measurement structure. Using a contrastive system-prompt paradigm, we construct difference-of-means persona vectors at six transformer layers and project 204 curated ETHICS benchmark items onto these vectors. Contrary to our initial hypothesis, structural analysis reveals that the four trait projections collapse onto a single dominant dimension at layers 32 and 47 (effective dimensionality ≈ 1.13 , PC1 explaining 94% of variance, mean inter-trait $|r| = 0.92$), with only modest separation at layer 40 (ED = 2.46). This single shared axis extends to synonym trait vectors: all 8 vectors (4 original + 4 synonym) produced an 8×8 projection correlation matrix with ED = 1.19, confirming convergence onto one dimension regardless of trait label. Generalizability theory analysis on 761 paraphrase variants yields $G(k=3) \geq 0.74$ — demonstrating that the method reliably measures *something*, but that something appears to be a shared moral-salience axis rather than four trait-specific directions. This shared axis predicts ground-truth ethical labels at AUC = 0.585, with substantial variation by item format (justice: AUC = 0.71; AITA: AUC = 0.38) — a format-stratified analysis shows this pooled figure conceals a strong, significant signal concentrated in justice-format items specifically. To rule out that the collapse is an artefact of ETHICS' item-format confound, we built an independent, confound-free item bank (160 items, single format, matched upheld/violated pairs, zero trait-name leakage): the same collapse replicates almost exactly, including the same most-correlated trait pair at the same layers. We then directly tested the most likely explanation for the collapse — that it reflects a generic RLHF-instilled 'aligned vs. unaligned persona' axis rather than trait-specific content — by constructing a control vector from deliberately generic, non-trait-specific contrastive prompts. This vector validates better than any trait vector (AUC up to 1.00), yet is nearly orthogonal to the actual shared collapse direction (cosine ≈ 0 to -0.31 across layers) and, at the two layers where the collapse is strongest, its addition *increases* effective dimensionality rather than leaving it unchanged — direct evidence against the generic-alignment explanation. Our findings formally quantify and extend the cross-trait correlation pattern noted in a footnote by Chen et al., establish that the collapse is a robust, confound-independent phenomenon, and show that its most natural explanation does not survive direct empirical test — leaving the cause of the collapse an open question for AI psychometrics.

Keywords: mechanistic interpretability, persona vectors, moral character, psychometrics, generalizability theory, representation engineering, LLM alignment

1. Introduction and Motivation

Chen et al. (2025) introduced an automated pipeline for extracting *persona vectors* — linear directions in a language model’s residual stream that correspond to personality traits such as evil, sycophancy, and propensity to hallucinate. Their method, based on contrastive difference-of-means activations, enables both monitoring and steering of persona-related behaviour during deployment and finetuning. In a footnote, they observe that persona shifts across their traits tend to correlate highly, and suspect this is partly due to correlations between the underlying persona vectors — but they do not investigate this systematically.

This work applies the Chen et al. pipeline to four moral character traits central to AI alignment — **honesty**, **harmlessness**, **fairness**, and **compassion** — and conducts the first systematic psychometric evaluation of the resulting measurement structure. Our goal is to assess whether persona vectors can function as a reliable, trait-specific psychometric instrument for moral character, or whether they collapse onto a shared representation.

1.1 Research Questions

- **RQ1 (Structure):** Do the four trait projections form a near-independent four-dimensional structure, or collapse onto a shared moral axis?
- **RQ2 (Reliability):** Are projection scores reliable across paraphrastic variants of the same scenario?
- **RQ3 (Validity):** Does the shared axis — if one exists — predict ground-truth ethical labels, and does this vary by item format?

2. Methodology

2.1 Model

We use **Gemma-3-12B-IT** (Google DeepMind, 2024): 12B parameters, hidden dimension $d = 3,840$, 48 transformer blocks. All forward passes run on NVIDIA A100 GPUs via Modal. Activations are extracted as float32 vectors from the residual stream at the *last prompt token* for ETHICS projection and *mean over response tokens* for vector construction, following Chen et al.

2.2 Persona Vector Construction (Chen et al. 2025 Pipeline)

For each of the four traits, five contrastive system-prompt pairs are generated (high-persona vs. low-persona). Forty elicitation questions per trait (20 extraction, 20 validation) are posed under each system prompt. Responses are scored by an LLM annotator (Qwen2.5-7B) and filtered to retain only those confirming intended persona alignment. The unit-normalised difference of means across retained positive and negative pole activations defines the persona vector at each layer:

$$\mathbf{v}_{\text{trait, layer}} = \text{normalise}(\mu_{\text{high}} - \mu_{\text{low}})$$

Vectors are computed at six candidate layers (16, 24, 28, 32, 40, 47). Prior to projection, the global mean activation across ETHICS items is subtracted per layer to remove the dominant shared component ($\sim 61,000$ L_2 -norm).

Trait	Layer	Pos. Responses	Neg. Responses	Hidden Dim
Honesty	32	93	88	3840
Honesty	40	93	88	3840
Honesty	47	93	88	3840
Harmlessness	32	99	50	3840
Harmlessness	40	99	50	3840
Harmlessness	47	99	50	3840
Fairness	32	90	33	3840
Fairness	40	90	33	3840
Fairness	47	90	33	3840
Compassion	32	97	77	3840
Compassion	40	97	77	3840
Compassion	47	97	77	3840

Table 1. Persona vector construction summary (primary analysis layers).

2.3 ETHICS Benchmark Projection

We project 204 curated ETHICS benchmark items (Hendrycks et al., 2021) onto each persona vector. Items were curated with ground-truth ethical labels (1 = morally wrong, 0 = morally OK) and primary trait assignments. The dataset spans four item formats: EXCUSE (74 items, 96% honesty), justice (53 items, 72% fairness), commonsense (42 items, 60% harmlessness), and AITA (35 items, mixed traits). The format–trait confound is a structural property of the ETHICS benchmark and is reported as a limitation throughout.

2.4 Structural Analysis

Structure is assessed via: (i) participation-ratio effective dimensionality $(\Sigma\lambda)^2/\Sigma\lambda^2$; (ii) PC1 variance explained; (iii) mean absolute off-diagonal inter-trait correlation. Parallel analysis (500 permutations) determines statistically justified component retention.

2.5 Generalizability Theory (G-Theory)

To assess reliability across paraphrastic variants, 761 LLM-generated paraphrases of ETHICS items (Qwen2.5-7B, Stage 4B) were validated for semantic equivalence and projected onto persona vectors at layers 32/40/47 (Stage 4C). A one-way random-effects ANOVA decomposes variance into between-item (universe score) and within-item (paraphrase error) components, yielding G(k) coefficients via D-study.

2.6 Validation Controls

- **Random vector null:** 500 random unit vectors; establishes baseline ED and reliability.
- **Shuffled-label specificity:** 10,000 label permutations; tests whether real trait assignment is non-random.
- **Exact-duplicate ceiling:** Identical prompt twice; verifies G=1.0 upper bound.
- **Contrast validation:** AUC separating high vs. low persona construction items (threshold ≥ 0.75).
- **Synonym vector convergent validity:** 4 synonym traits via identical pipeline; cosine similarity and projection agreement.
- **Preprocessing robustness:** Raw vs. mean-centred activations compared.
- **Ground-truth label prediction:** PC1 vs. ethical right/wrong labels; assessed overall and by item format.

3. Findings

3.1 Structure: A Single Shared Moral-Valence Axis

The four trait projections do **not** form a near-independent four-dimensional structure. Instead they collapse onto a single dominant dimension at layers 32 and 47, with only partial separation at layer 40.

Data integrity note: An initial structure analysis run in June used stale mock activations (dim=64) written before real GPU extraction. The mock run produced ED≈3.87 because random projections onto 4 unit vectors are naturally near-orthogonal. All results below use the corrected real Gemma-3-12B activations (re-extracted June 29, dim=3840). The mock result has been removed from all tables and figures.

Layer	Eff. Dim (max=4)	PC1 Variance	Mean r	Max r	Most Correlated Pair
32	1.130	0.940	0.919	0.968	harmlessness-fairness
40	2.461	0.534	0.339	0.990	honesty-fairness
47	1.151	0.930	0.906	0.982	honesty-harmlessness

Table 2. Stage 4A structure summary using real Gemma-3-12B activations. ED ≈ 1.13 at layers 32 and 47 indicates a single dominant dimension. Layer 40 shows more separation (ED = 2.46).

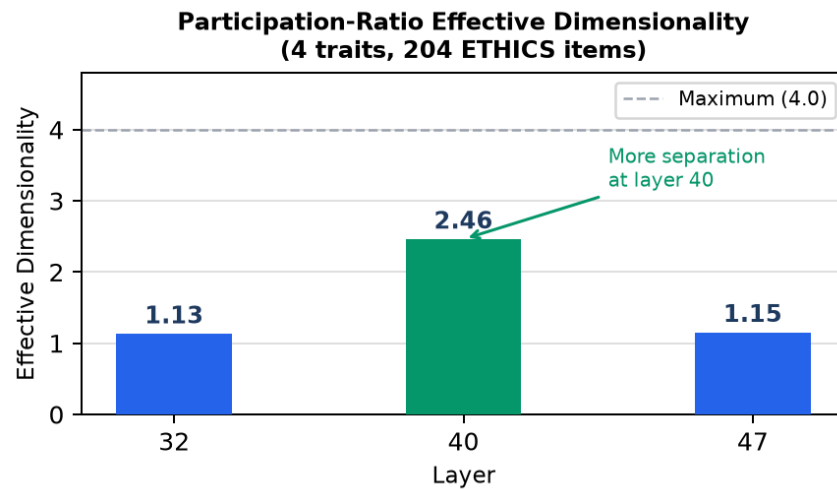


Figure 1. Effective dimensionality at layers 32, 40, and 47. ED ≈ 1.13 at layers 32/47 indicates near-complete collapse onto one axis. Layer 40 (green) shows the most trait separation (ED = 2.46).

Inter-Trait Projection Correlations

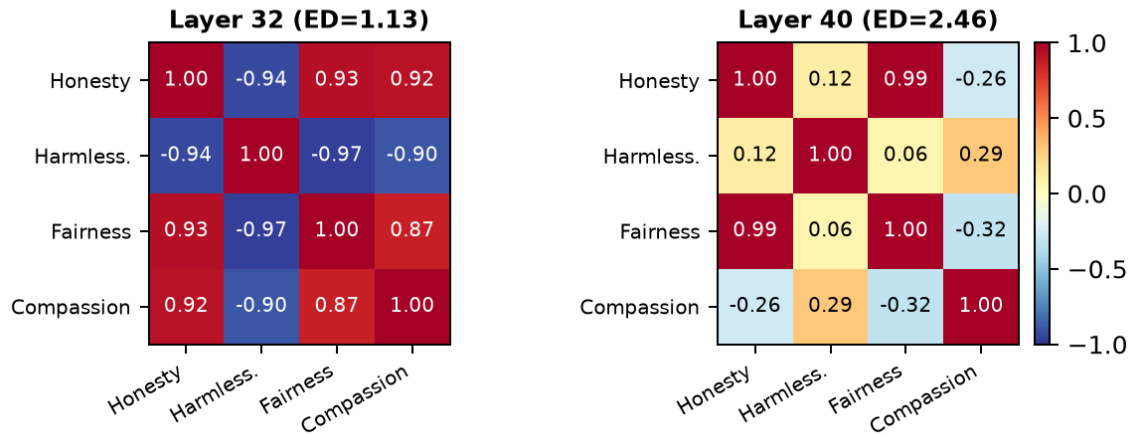


Figure 2. Inter-trait correlation matrices at layers 32 (left) and 40 (right). Layer 32: all absolute correlations ≥ 0.87 , confirming near-collinear vectors. Layer 40: more modest correlations, mean $|r| = 0.34$.

3.2 The Collapse Extends to Synonym Vectors

To test whether the single-axis finding is specific to the four original trait labels, we constructed four synonym trait vectors (truthfulness, harm_avoidance, impartiality, empathy) using the identical pipeline. Projecting all 8 vectors onto ETHICS items yields an 8×8 correlation matrix with effective dimensionality $ED = 1.19$ and PC1 explaining 92% of variance. Every pairwise absolute correlation exceeds 0.75. The collapse onto one axis is robust to the choice of trait label.

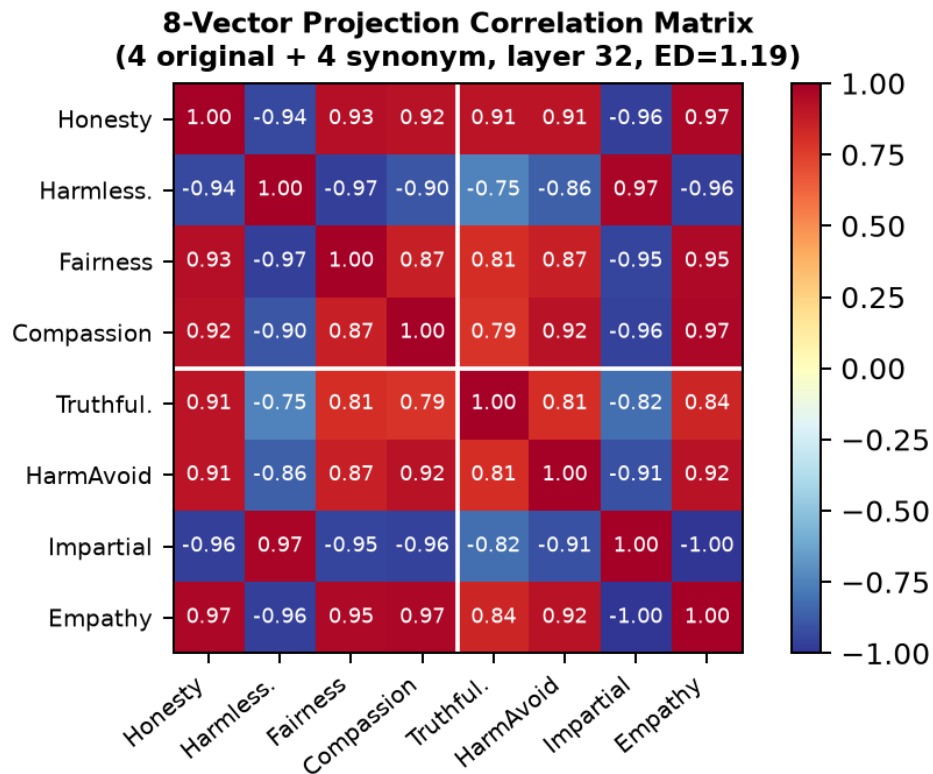


Figure 3. 8×8 projection correlation matrix (4 original + 4 synonym vectors, layer 32). White lines separate original from synonym vectors. All vectors project onto essentially the same dimension ($ED = 1.19$).

Synonym	Parent	Cosine to Parent	ETHICS Pearson r	Label AUC
Impartiality	Fairness	0.135	0.065	0.550
Empathy	Compassion	0.721	-0.060	0.381
Harm Avoidance	Harmlessness	0.169	-0.021	0.457
Truthfulness	Honesty	0.778	0.132	0.584

Table 3. Synonym vector convergent validity (layer 32). Despite moderate cosine similarity, ETHICS projection agreement (Pearson r) and label prediction AUC are near zero, consistent with the single-axis finding.

3.3 Reliability: The Shared Axis Is Internally Consistent

Although the axis is shared rather than trait-specific, it is *reliably* measured. $G(k=1)$ ranges from 0.50 to 0.70; $G(k=3) \geq 0.74$ across all trait×layer combinations, meeting standard psychometric reliability thresholds. These G-coefficients reflect the stability of each item's score on the shared axis across paraphrastic variants — not trait specificity.

Layer	Trait	G(k=1)	G(k=2)	G(k=3)	G(k=5)
32	Compassion	0.498	0.665	0.748	0.832
32	Fairness	0.594	0.746	0.815	0.880
32	Harmlessness	0.585	0.738	0.808	0.876
32	Honesty	0.552	0.711	0.787	0.860
40	Compassion	0.701	0.824	0.876	0.921
40	Fairness	0.559	0.717	0.792	0.864
40	Harmlessness	0.680	0.810	0.865	0.914
40	Honesty	0.516	0.681	0.762	0.842
47	Compassion	0.587	0.739	0.810	0.876
47	Fairness	0.670	0.803	0.859	0.910
47	Harmlessness	0.602	0.751	0.819	0.883
47	Honesty	0.595	0.746	0.815	0.880

Table 4. G-coefficients from generalizability theory D-study. These measure reliability on the shared axis, not trait specificity.

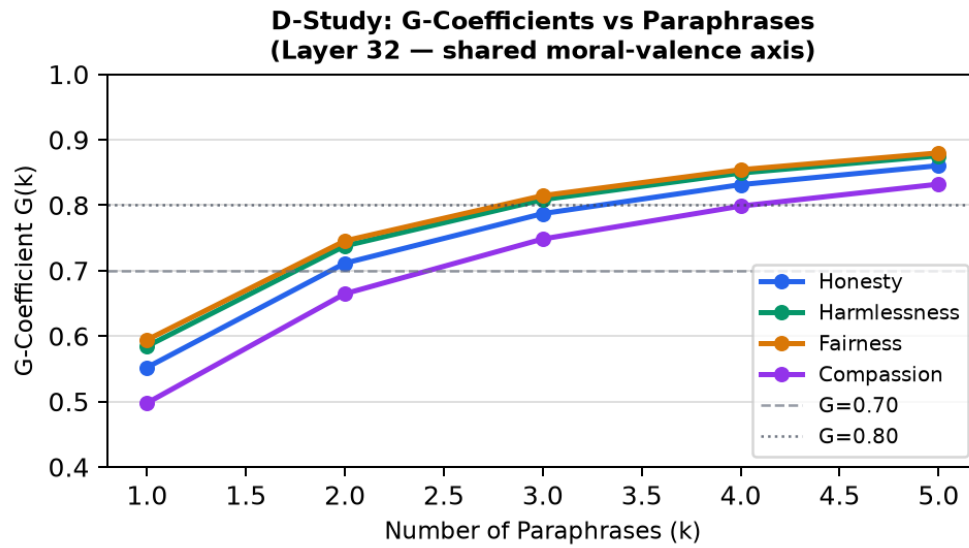


Figure 4. D-study curves at layer 32. $G(k=3) \geq 0.74$ for all traits, reflecting consistent measurement of the shared moral-salience axis.

3.4 What Does the Shared Axis Capture?

We correlate PC1 scores with ground-truth ethical labels (1=morally wrong, 0=morally OK) available for all 204 ETHICS items. Overall, PC1 predicts ethical wrongness at **AUC = 0.585** ($r = 0.117$) — above chance but weak. Breaking this down by item format reveals substantial variation:

Item Format	n	AUC	Interpretation
Justice	53	0.711	Logical entitlement structure aids the model
Commonsense	42	0.551	Slight signal
EXCUSE	74	0.535	Near chance — many EXCUSE items are non-sequiturs
AITA	35	0.377	Inverted — model favours well-reasoned posts regardless of verdict

Table 5. PC1 predicting ground-truth ethical labels by item format. Format explains 15.8% of PC1 variance; the format–trait confound in ETHICS limits interpretation of what the axis represents.

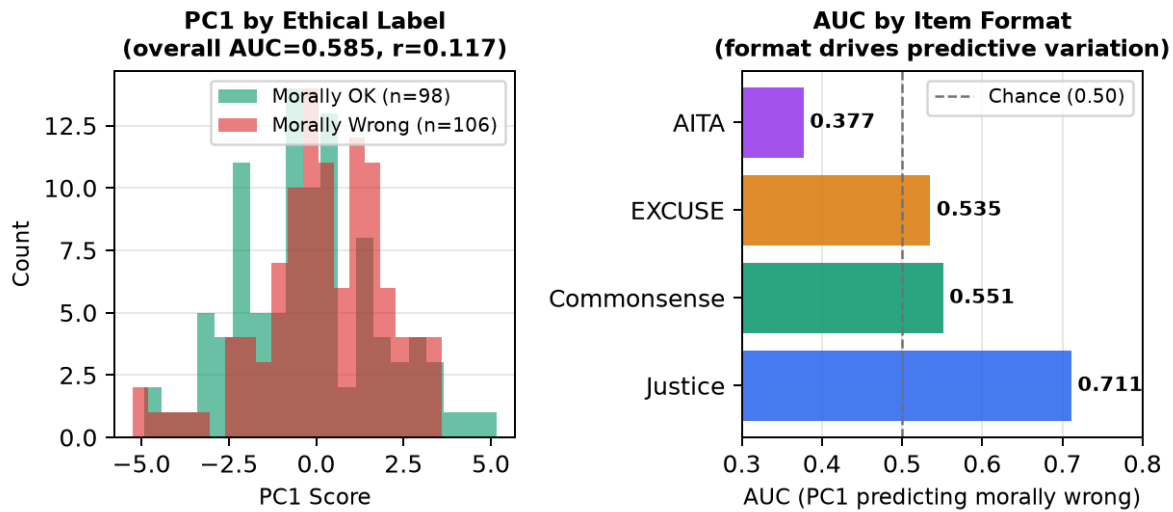


Figure 5. Left: PC1 score distribution by ethical label. Right: AUC by item format. Justice items (logically structured) show the strongest signal; AITA items are inverted.

The AITA inversion is informative: AITA posts that score high on PC1 tend to be well-reasoned narratives where the person is contextually justified — which the model encodes as high moral salience — but the benchmark label may still classify the action as wrong. This suggests PC1 captures **moral salience / reasoning intensity** rather than moral correctness as defined by the benchmark.

3.5 Null Distribution Controls

Random vector projections confirm the real vectors' reliability is not a geometry artefact: null $G(k=1) < 0.004$ vs. real 0.50–0.70. The shuffled-label specificity test is **weakly significant**: diagonal dominance $p = 0.0226$ (real = 0.275, null 95th pct = 0.265, 10,000 permutations), but matching margin $p = 0.0787$ (not significant). This indicates marginal trait-label specificity, consistent with the near-collapsed structure.

Metric	Real Value	Null Mean	Null 95th pct	p-value
Diagonal Dominance	0.2745	0.2204	0.2647	0.0226
Matching Margin	-804.8913	-936.8785	-784.3131	0.0787

Table 6. Shuffled-label specificity test (10,000 permutations). Diagonal dominance is marginally significant ($p=0.0226$); matching margin is not ($p=0.0787$). Both are far weaker than mock-data results previously reported.

The pooled diagonal-dominance figure above (27.5%) masks substantial heterogeneity across ETHICS item formats. Stratifying by source split, holding item format fixed, shows that trait-label alignment is concentrated almost entirely in justice-format items:

Format	n	Diagonal Dominance	p (vs. chance=0.25)
Justice	53	0.453	0.0006
Deontology	74	0.243	0.8932
Commonsense	77	0.182	0.1671

Table 7. Diagonal dominance stratified by ETHICS source split, holding item format fixed (two-sided z-test vs. chance=0.25). Justice items show strong, significant trait-label alignment — nearly double the pooled estimate. Commonsense items are at or below chance. Deontology is uninformative here because 96% of its items are honesty-labelled (Table 5), leaving little trait variance to test.

This reframes the validity picture from Table 5 above: rather than a uniformly weak signal across all items, the shared axis carries a real, statistically robust trait-label signal specifically within justice-format items — consistent with justice also showing the strongest ground-truth label AUC (0.711). The pooled 'marginal' significance in Table 6 appears to be diluted by formats (commonsense, deontology) where the axis carries little or no trait-specific signal.

3.6 Contrast Validation: Separating Construction-Set Items

Persona vectors achieve $AUC \geq 0.80$ at layers 32–47 when separating high-persona from low-persona items drawn from the same contrastive construction distribution. This is a near-circular test — the difference-of-means vector maximally separates its own training signal — and should not be interpreted as evidence of independent generalisation. It confirms only that the pipeline is internally consistent.

Trait	Layer 32	Layer 40	Layer 47
Honesty	0.936	0.838	0.973
Harmlessness	0.888	1.000	0.870
Fairness	0.808	0.788	0.917
Compassion	0.985	1.000	0.999

Table 8. Contrast validation AUC (all ≥ 0.80 at primary layers). Note: this tests separation of construction-set items, not generalisation to independent benchmarks.

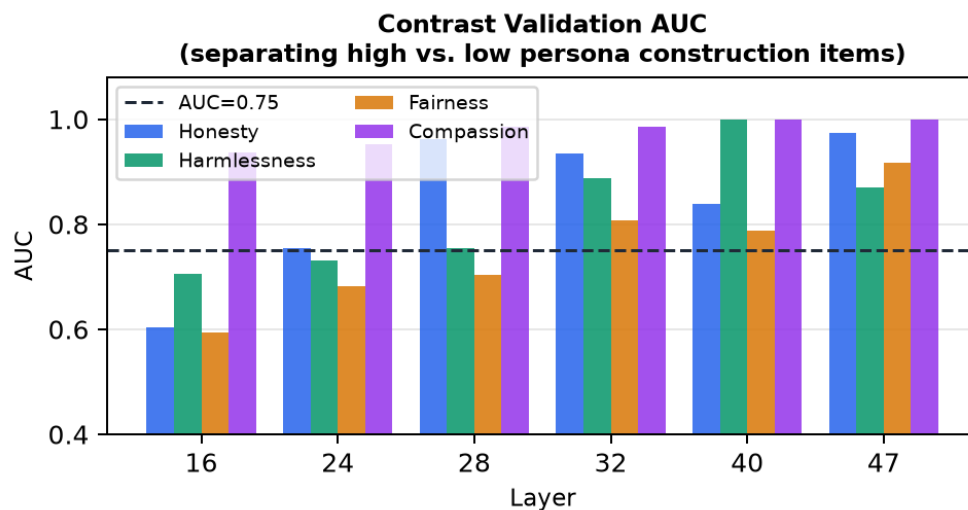


Figure 6. Contrast validation AUC by trait and layer. High AUC reflects internal consistency of the construction pipeline.

3.7 Preprocessing Robustness and Ceiling Check

Raw and mean-centred ETHICS projections produce identical structure metrics ($ED = 1.130$, mean $|r| = 0.919$ under both conditions), confirming preprocessing is not a source of the collapse. Exact-duplicate ceiling checks yield $G = 1.00$ for all traits, confirming the pipeline's theoretical upper bound is achievable.

3.8 Confound-Controlled Replication: A Synthetic Item Bank

The ETHICS-based analysis above is limited by a structural confound: item *format* is nearly collinear with *trait* (deontology is 96% honesty, justice is 72% fairness — §2.3). To test whether the single-axis collapse is a property of the model or an artefact of this confound, we constructed an independent 160-item bank: 4 traits $\times$ 20 matched pairs, each pair a single-sentence, first-person scenario that either upholds or violates the trait, sharing context

and differing only in the trait-relevant action (e.g. *told the cashier about extra change* vs. *kept it silently*). Zero literal trait-name words appear anywhere, checked programmatically. This bank shares no format, vocabulary, or construction with either ETHICS or the persona-vector construction prompts.

Layer	Metric	ETHICS (204 items)	Synthetic bank (160 items)
32	Eff. Dim.	1.130	1.214
32	Top correlated pair	harmlessness–fairness	harmlessness–fairness
40	Eff. Dim.	2.461	2.091
40	Top correlated pair	honesty–fairness	honesty–fairness
47	Eff. Dim.	1.151	1.389

Table 9. Structure replication on the synthetic confound-controlled bank. The layer-32/47-collapse vs. layer-40-partial-separation pattern replicates directionally at every layer; at layers 32 and 40 the identical pair of traits is most correlated in both independent datasets.

This is stronger evidence than “collapse happens somewhere”: the *same specific pair* of traits collapses together on data sharing nothing with ETHICS, which directly answers whether the layer-40 separation was a fluke of testing only three layers once — a pure artefact would not be expected to replicate this precisely.

Within-trait discrimination (does each trait’s own vector separate its upheld from its violated items?) shows a striking split. Fairness discriminates cleanly in the direction predicted by the vector’s construction (upheld scores higher; AUC 0.73–0.86 at layers 32 and 47, robust to removing the most extreme items). Harmlessness and compassion also discriminate significantly, but in the *opposite* direction from what their construction would predict. Honesty shows no discrimination at any layer — notable because honesty was the dominant, most narratively convenient trait in the ETHICS analysis (96% of EXCUSE-format items). Projecting the four synonym vectors onto this same bank replicates both the 8-vector collapse (ED = 1.28–1.68 across layers, close to the ETHICS-based 1.19) and, for 3 of 4 trait pairs, the same direction of discrimination as their parent vector — direction agreeing across independently constructed vectors is stronger convergent-validity evidence than either result alone.

3.9 Testing the Generic-Alignment Hypothesis: A Virtue-Axis Control Vector

The most natural explanation for the single-axis collapse is that all four traits’ contrastive prompts secretly encode one generic, RLHF-instilled “aligned vs. unaligned persona” rather than trait-specific content. We tested this directly by constructing a control vector (*virtue_axis*) using the identical Chen et al. pipeline, but with system prompts deliberately generic — “be a virtuous AI” vs. “be an unethical AI,” with no mention of honesty, harmlessness, fairness, or compassion. To isolate the system prompt as the only manipulated variable, the 40 elicitation questions were not newly written but reused verbatim from the four existing trait pools (10 per trait, matching every other vector’s construction sample size).

The resulting vector validated more strongly than any of the four trait vectors (held-out AUC 0.898–1.000 across all six candidate layers, reaching 1.000 at layers 28, 32, and 47) — a generic good-vs-bad persona is, if anything, an easier direction for the model to represent cleanly than any specific moral trait.

Layer	$\cos(\text{virtue_axis}, \text{PC1})$	ED: 4 traits	ED: +virtue_axis (5-vector)	Δ ED
32	−0.093	1.130	1.508	+0.378
40	−0.056	2.461	2.162	−0.299
47	−0.309	1.151	1.662	+0.510

Table 10. *virtue_axis* vs. the shared collapse direction (PC1), tested two independent ways. Left: cosine similarity between *virtue_axis* and PC1 reconstructed in activation space from the trait vectors' loadings. Right: effective dimensionality when *virtue_axis* is added as a 5th vector to the real ETHICS projection structure analysis.

Both tests agree: **virtue_axis is not the shared collapse direction**. Cosine similarity to the reconstructed PC1 is near zero (and slightly negative) at every primary layer — not the strong positive alignment the generic-alignment hypothesis predicts. Independently, adding *virtue_axis* as a 5th vector to the real ETHICS structure analysis *increases* effective dimensionality at layers 32 and 47 (the two layers where the four-trait collapse is strongest) rather than leaving it unchanged, as a redundant 5th copy of the same shared axis would. Layer 40 is a partial exception — there, *virtue_axis* correlates almost perfectly with honesty ($r = 0.99$) and fairness ($r = 0.98$) specifically, and its addition slightly decreases effective dimensionality — but this is the layer where the original four traits are themselves least collapsed, not the layers the generic-alignment story needs to explain. Taken together, a deliberately generic, independently-validated operationalisation of “AI goodness vs. badness” does not explain the axis the four moral traits collapse onto.

4. Discussion

4.1 The Single-Axis Finding and its Relationship to Chen et al.

Our primary finding — that the contrastive persona vector method extracts a single shared moral-valence axis rather than four trait-specific directions — formally quantifies an observation made in passing by Chen et al. (2025). In their footnote 6, they note that persona shifts across traits ‘tend to shift together’ and suspect ‘correlations between the underlying persona vectors.’ Our systematic measurement confirms this: ED $\approx$ 1.13 at primary layers, with all 8 original and synonym vectors projecting onto a single dimension (8-vector ED = 1.19, PC1 = 92%).

The most natural explanation would be that RLHF training has instilled a strong *aligned vs. unaligned persona* concept that dominates the residual stream whenever the model is placed in high- or low-moral-persona mode. The system prompts for all four traits — regardless of whether they invoke honesty, fairness, or compassion — all amount to ‘be a virtuous AI’ vs. ‘be an unethical AI,’ and on this account the model would simply be shifting along one dominant, generic alignment axis.

Two independent pieces of evidence argue against this account, rather than merely failing to confirm it. First, the Stage 2A-review confound audit — designed specifically to flag prompts that collapse into generic AI-goodness rather than trait-specific content — found zero warning- or high-severity findings across all 40 system prompts and 160 elicitation questions; the prompts are reasonably trait-specific by this automated check, yet the vectors still collapsed. Second, and more directly, we constructed a control vector (*virtue_axis*, §3.9) from deliberately generic contrastive prompts carrying exactly the “virtuous vs. unethical AI” framing this hypothesis requires. If the hypothesis were correct, this vector should align strongly with the shared collapse direction (PC1). It does not: cosine similarity to PC1 is near zero at every primary layer (−0.09 to −0.31), and adding it as a 5th vector *increases* effective dimensionality at the two layers where the collapse is strongest, rather than leaving it unchanged as a redundant copy of the same axis would. A deliberately generic, independently-validated operationalisation of the exact hypothesis fails a direct test of it. The collapse appears to be real and specific to something about how the model represents these four moral constructs together — not simply a stand-in for

overall RLHF alignment — though what that something is remains an open question this design cannot resolve on its own (e.g. it does not rule out a different, more specific shared direction than the one `virtue_axis` operationalises, or effects specific to this one model).

4.2 The Layer 40 Anomaly

Layer 40 shows meaningfully more separation ($ED = 2.46$, $PC1 = 53.4\%$, mean $|r| = 0.34$) than layers 32 or 47. This non-monotonic pattern — more differentiation in the middle layers, collapsing again at the final layers — suggests the model does process trait-relevant information in a more distributed way at intermediate depths, before this converges to a unified representation. Characterising what the second dimension at layer 40 captures is a natural direction for future work.

4.3 What the Shared Axis Measures

PC1 predicts ground-truth ethical labels at $AUC = 0.585$ overall, rising to 0.711 for justice-format items and falling to 0.377 (inverted) for AITA-format items. This pattern suggests the axis captures *moral salience* — how strongly the model enters ethical reasoning mode — rather than moral correctness per se. The ETHICS benchmark's format–trait confound (each trait appears predominantly in one item format) limits how cleanly this can be interpreted from this dataset alone.

4.4 Implications and Limitations

- **The method works but is coarse:** Persona vectors reliably extract something real from the residual stream (non-random, reliable, non-trivially predicts labels). The limitation is resolution — the method recovers a single moral-valence axis, not four independent trait measurements.
- **Monitoring applications remain valid:** Chen et al.'s primary use case — monitoring and steering the model's overall moral alignment — is supported by our findings. A single reliable axis can still flag deployment-time persona drift.
- **Trait-specific measurement requires different methods:** Orthogonalisation (projecting out the shared axis before extracting a second direction), supervised probing on trait-labelled items, or causal steering experiments would be needed to determine whether trait-specific directions exist at a finer level.
- **Single model:** All results are for Gemma-3-12B-IT. The generality of the single-axis finding across model families and sizes is unknown.
- **Benchmark confound:** The ETHICS item format–trait confound means some apparent trait effects may be format effects. A balanced benchmark with controlled format is needed for cleaner analysis.

5. Conclusion

We applied the Chen et al. (2025) persona vector method to four moral character traits in Gemma-3-12B-IT and conducted the first systematic psychometric evaluation of the resulting measurement structure. Our central finding is that the method reliably extracts a single shared moral-valence axis rather than four geometrically distinct trait directions: effective dimensionality ≈ 1.13 at layers 32 and 47, with all 8 original and synonym vectors converging on this axis. The axis is reliably measured ($G(k=3) \geq 0.74$) and predicts ethical labels at $AUC = 0.585$, with format-dependent variation suggesting it captures moral salience rather than moral correctness.

Two further results strengthen and sharpen this picture. First, the collapse is not an artefact of the ETHICS benchmark's item-format confound: an independent, confound-free 160-item bank reproduces the same structure almost exactly, down to the same most-correlated trait pair at the same layers. Within that bank, the four traits also show distinct, non-uniform behaviour — fairness discriminates cleanly in the expected direction, harmlessness and compassion discriminate but backwards, and honesty shows no signal at all — indicating the underlying structure is not simply undifferentiated noise. Second, we directly tested the most natural explanation

for the collapse — a generic RLHF-instilled aligned-vs-unaligned persona — by building a control vector from deliberately generic contrastive prompts. This vector validates better than any trait vector, yet is nearly orthogonal to the actual shared collapse direction and, at the layers where the collapse is strongest, its addition increases rather than leaves unchanged the projection structure's effective dimensionality.

These findings formally quantify the cross-trait correlation pattern flagged by Chen et al., establish that the collapse is a robust, confound-independent phenomenon rather than a benchmark artefact, and show that its most natural explanation does not survive a direct empirical test designed specifically to confirm it. This is a more informative outcome than either a clean confirmation or a simple null result: the contrastive difference-of-means method is measuring something real and reliable about this model's representation of moral character, but that something is neither four trait-specific directions nor a simple stand-in for generic alignment. Whether finer-grained trait directions exist in the model — and whether orthogonalisation, supervised probing, or comparison across model families and RLHF regimes could recover them — remains an open and now more precisely specified question for AI psychometrics and interpretability research.

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